

(1.) Based upon your experiences as a Chemical Engineering student at UIC, please rate your skills in the following categories, using the following criteria:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

- 1.50 Ability to apply knowledge of mathematics, science and engineering.
- 1.63 Ability to design and conduct experiments, as well as to analyze and interpret data.
- 1.67 Ability to design a system, component, or process to meet desired needs.
- 1.39 Ability to function on multi-disciplinary teams.
- 1.50 Ability to identify, formulate, and solve engineering problems.
- 1.26 Understanding of professional and ethical responsibility.
- 1.45 Ability to communicate effectively.
- 1.46 Broad education necessary to understand the impact of engineering solutions in a global and societal context.
- 1.40 Recognition of the need for, and an ability to engage in life-long learning.
- 1.85 Knowledge of contemporary issues.
- 1.57 Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.
- 1.73 Overall, do you feel that your undergraduate degree will prepare you for your career?

(2.) Which courses or topics were the most interesting or useful?

Aspen, Senior design

Transport Phenomena I, II and III, Senior Design

chemistry classes

CHE 312, 313, 321, 396

The most interesting topics I found to be the 300 level courses. I enjoyed the technical nature of them.

The most important were definitely Transport Phenomena - although I feel like I had to cover a lot of ground in Transport II with Sharma because Transport I Caracotsios taught a completely different curriculum.

I liked learning about various career paths that people took.

Aspen Simulation. Numerical Methods. Senior Design. Unit Ops Lab. Transport Phenomena III. Transport Phenomena I. Transport Phenomena II. CS 109.

Thermodynamics 1, CHE 311, CHE 313, CHE 321, Senior Design, Senior Design Lab

The topic that i thought to be the most interesting was CHE311

che 312, che 321

Heat and Mass Transfer and Reaction Engineering

210, 321, and 396

CHE 311, CHE 312, CHE 313, CHE 433

heat and mass transfer, renewable energy technologies, material and energy balancees

311, 312, 433

Transport phenomenon was the most interesting and had the biggest impact on me

ChE Lab, Transport Phenomena, Material & Energy Balances

Transport phenomena (all 3)

I liked fluid mechanics and aspen

Overall, CHE 311 (Transport Phenomena) was interesting and has been extremely useful in helping me understand topics that are difficult to completely grasp the first time around. All the thermo classes have also been useful.

Thermodynamics and Fluid dynamics. Che 396 was also interesting but it had way too much work/ way too much was expected of us.

312, 321, 210, 205, 396

Reaction Engineering, Senior Design

The most interesting were material and energy balances, thermodynamics, and transport.

Thermodynamics, labs, aspen, and senior design

Transport Phenomena 1,2,3; Che 205 Numerical Methods; Che 396; Che 433

che 433

useful: senior design, thermodynamics, engineering lab, mathematical tools for chemical engineers Interesting: transport phenomena, biochemical engineering

Both thermo courses were pretty good. I also enjoyed CHE321

Aspen

CHE 313, CHE 321, CHE 433 CHE 312

Chemistry was the catalyst to get me into this major. Especially organic. CHE 321 was also very interesting.

Senior design

all courses are useful and are useful for the future.

CHE312, CHE205, CHE210

Material and Balances, CHE 313 separation, senior design, heat and mass transfer, fluid mechanics, thermodynamics

Mass and energy balance and Aspen simulation

396 topics, thermo, and reactions were great

ChE 381 and 382 were extremely useful for understanding how the equipment we learned about in other course functioned in the real world

ChE 433 ChE 312 ChE 321

transport 1,2,3 aspen

311 and 396

macroscopic heat transfer. che 210. senior design

senior design i feel will be the most useful

301 312 205

CHE 433 and CHE 321 were my favorite classes.

All Transport Phenomenon Classes

Thermodynamics Design Lab

N/A

CHE 301, 311, 312

transport sequence, thermodynamics

The required mathematics classes were most useful, and are incredibly necessary. Following those classes the most interesting courses would be transport phenomena I & II as well as thermodynamics

(3.) Which courses or topics were the least interesting or useful?

N/A

CHE 205, CHE 321

ECE 210

Physical chemistry II

Anything that we had to take outside the spectrum of Math and Science.

All of them were interesting.

Quantum Chemistry. ECE 210. Physical Chemistry I (very redundant with Thermodynamics)

Physical Chemistry 2- Give me a break. It was so disruptive to my education. Everyone just memorized things the professor did and wrote it down on the exams (averages in the 30s!). What a waste of time!

The last i thought was least interesting/hard to understand was CHE321

che 313

None

Aspen

CHE 205

che 313

205, 313

CHE 205, I hardly used the skills

Physical Chemistry II is the only course where I have not been interested in the material in and of itself.

ECE 210 and organic chemistry

i hated the course 205 and analytical chemistry

For me, CHE 381 (Unit Ops lab) was not particularly interesting. The course is extremely useful in that this is the first time we can do anything hands-on with the concepts we've learned. However, the experiments were not very exciting. Something more innovative, like looking at solar cells or newer technology would have made the class a lot more interesting.

Physical chemistry 2

313

205 was useful at the time I suppose but in retrospect only interpolation/extrapolation methods and use of VBA ever came in handy in the end.

The least interesting were separations and senior design. Physical chemistry is not useful.

CME

Che 321 Reaction Engineering was the least interesting to me

che 210

I really liked all my courses. but material and energy balances felt like we were not learning much.

CHE 396 was pretty difficult to get through as the lectures were completely uninteresting and the projects made unnecessarily difficult by random assignment.

Pchem

NONE

Physical chemistry, analytical chemistry (very small use gained for extreme amount of effort), CS109 should be prerequisite for CHE 205.

Pchem 2

least interesting for me was CHE 205. Useful but not an interesting class

CHE381, CHE201, Physical Chemistry II

205-computational methods

Physical chemistry II, Analytical chemistry

312

None

P. Chem I P. Chem II

reaction engineering

205

gen eds :P but all of the engineering we did is great exposure to the different kinds of problems that we can solve using the engineering mindset

liquid transport

201 311

CHE 205 was not interesting to me.

The Computational Methods/VBA Class

Mass Flow

N/A

CHE 205, a more integrated course with in-depth analysis of numerical methods

che 201, physical chemistry

CHE 205 was not very helpful and incredibly boring. I learned an equal amount if not more about C++, VBA, and Matlab in CS 109 simply without specific chemical engineering applications.

(4.) What courses/topics should we consider deleting or adding to the curriculum?

Aspen

I would place CS 109 as a pre-requisite for CHE 205.

remove Physical Chemistry II, and add an additinal course for chemical reaction engineering.

I feel CHE 433 should be a required course. The manner in which it is taught is very useful for practical engineering. You feel you know chemical engineering and when you get to this course you realize you know very little when it comes to designing process equipment. Chem 346 could be eliminated from the undergraduate curriculum and replaced with something more applicable to undergraduate engineers.

Add more on to CHE205, I felt like that class was a really good class but it needed to be taught junior year or made easier for incoming students.

Talk about how people ended up getting the job, like if it was online, in person, etc.

Delete Quantum. It is a ridiculous requirement. Please replace it with statistics, stoichastic calculus, or linear algebra.

Delete Physical Chemistry. Make Aspen a requirement.

Add more computer science related courses on how to code in CHEME

delete che 205

None

Linear Algebra add

Numerical methods

add an ethics course. this is needed.

Better numerical methods course, add linear to curriculum

Maybe add Aspen as mandatory

I would suggest: - Making CHE 433 a required course (if it's an essential component of everyone's senior design projects, to the point where one of the team assembly criteria is that every team must have someone who took Aspen, then it ought to be mandatory) - Condensing Physical Chemistry into one course for chemical engineers specifically (I don't see how an entire semester of quantum physics is going to be useful in industry - a background in it would seem to be sufficient) - Using the course slot freed up by condensing PChem into one semester for the technical elective (I wanted to take something else as a technical elective initially, but with the senior design project requiring knowledge of simulations, it didn't really feel like I could elect to take anything else without paying the price)

ECE 210 can be removed

delete analytical chemistry and pchem 2

Although not all courses/topics were interesting, I feel that everything we have covered is essential to understanding the various problems we will have to deal with as professionals so I would not delete anything from the curriculum.

Physical Chemistry 2 - its pointless

Computational Methods, but with regard to real engineering problems.

I would consider redesigning the senior design course. I think physical chemistry 1 should not be required as part of the curriculum since chemical engineers take two thermodynamics classes already and the class is poorly taught. Physical chemistry 2 should not be required since chemical engineers do not often deal with quantum systems.

delete CME and add aspen as a requirement

None

none

Add: environmental engineering issues.

I am not sure why pchem II is a required course. I feel like I could have spent that slot taking a CHE elective and gained much more from that

Remove Pchem2

None

Get rid of physical chemistry. We have enough thermodynamics classes and quantum is not needed in industry. Replace both classes with more electives so we can gain more knowledge to our major.

none

Physical Chemistry II seems kind of useless, I think Linear Algebra would prepare us a lot better

Consider deleting or offering the 205 computational methods course as a 300 or 400 level course.

Delete pchem II and add more ME related classes.

delete PChem 2 as a requirment

More on programs and other tools utilize in industry

Removal of physical chemistry Addition of more technical electives for the various disciplines that a ChemE could enter

delete physical chemistry

205 is a class where the material is never applied in other courses

add ASPEN to the required because you have that program being used in other classes

more physics, statics

Maybe add a statistics course

The Aspen Class as mandatory

N/A

I think a history course on chemical engineering would be stellar and, instead of engineering 100, a comprehensive course on engineering reports. This could be one combined course with the English/History department.

physical chemistry 1 and 2

Delete computational methods of chem e. In it's place add a matlab/vba coding requirement, a lot of students need a refresher course.

(5.) Do you have any additional comments or suggestions?

None

I really enjoyed 205; however, I was not ready for that course my sophomore year.

No

Get rid of quantum.

No

N/A

no

None

no

senior design needs group evaluation rubric. senior design needs actual team guidelines and a professor willing to look into group dynamics and help students be better in the workplace. very imperative.

No

No

If the department is moving into another building, the new building ought to have more computers, not less, especially if there are going to be more engineering students - the number of computers as it currently stands is barely sufficient. Or alternately, use another simulation software that is made by a company that doesn't make it impossible for students to have temporary home copies for the duration of their senior year (if such a company exists - I don't actually know, but if Matlab can do it, so can UniSim or the like.) Also, it would be much better if CHE 311 and 312 were both taught by the same professor in consecutive semesters, since there is a high level of continuity between the two courses.

Some way to incorporate real world news/issues

none

Buy more working computers.

no

Che 433 Professor should be different

no

This is a very strong program. I learned a lot.

Working with some classmates in 396 and 381 has me deeply concerned about some of the students that will apparently be graduated with a degree. While there seemed to be a lack of motivation to do work, there was also an alarming lack of what should be basic knowledge by now for a few individuals.

none

no

I don't think CHE201 prepared me well for CHE301

I enjoyed my experience as an undergraduate in the chemical engineering department.

None

n/a

None

no

no

We have a really great group of professors teaching the program. Wedgewood, Sharma, and Zdunek in particular really made class fun and provoking of my curiosity which is the probably the best thing professors could do for their students. I'm glad to have learned from them!

none

No

N/A

N/A

Greater collaboration amongst professors on scheduling exams. Consistency within 311 and 312 course teaching.

none

The college of engineering makes it very difficult to get credit for other general electives that transfer students got from other schools which is annoying when they do not pertain to their degree at all. I for example, had to retake english II because my appeal for credit was denied.

(6.) How many years/months did it take for you to complete your B.S. degree in chemical engineering?

5.5 years

4 years at UIC. 1 year at Loyola under a different major.

4 years

4 years

4 years

4 years

5 years.

6 years (I transferred from a city college)

4.5 years

4 years

4 years

5 years

5 years

8 semesters

January 2015 - May 2017 (came in with transfer credits)

4.5 years

5 years

4 years

3 years

4 years

4 years

5 years

5 years

3.5

3 years

4.5 years

5 years in total

About 6 1/2 years to complete my A.S. and B.S. with a small gap in the middle.

4 (+ will take an extra semester next year)

5 years

4 yrs

2.5 + 2 years community college

5 years

4 Years

5. I worked all throughout school

5 years, I changed from chemistry and transferred to UIC starting in year 3.

N/A

4 years/ 40 months

4 years

It took 5 years. A little long, and for transfer students that is pretty much how long it will always take. Try to fix that because students are poor and debt seems to last forever.

(7.) Did you work part-time while you were a student? If yes, then approximately how many hours per week?

yes, about 15-18 hours per week

Yes, at least 20.

yes. hours vary. probably 5 towards the end, but 7-10 in the begining

I did not work while I was a student.

Yes, 20.

Yes. About 20-23 hours a week.

Only years 2 & 3. Approximately 20.

Yes. 25 hours/week

Yes, 20 hours a week.

no

Yes, 20 hours per week.

Intermittently

yes, about 12

yes. sometimes 16 hours.

Yes, 15

yes

Yes. about 20 hours a week

yes and approximately 15 hours a week

Yes. Approximately 20 hours per week.

Yes, 5-13 hours.

yes, about 30 hours/week

Yes, 10 hours.

No

Yes, sometimes full time, sometimes not at all, sometimes part time. 0-40 hrs a week

no

no

Yes, first year for 24 hours, second year for 18 hours, first half of third year for 8 hours. last year, did not work.

Yes, 20-30 hours a week.

Yes

Yes i worked part time. 17 hours a week

Not while I was at UIC

Yes, about 20 hrs

no

Yes, anywhere between 5-30 hours per week

Yes, about 20 to 30 hours per week.

No

no

For 7 months of my 3rd year. About 20 hours/week

Yes, 20-40 (varied)

yes first 2 years i worked 40 35 hrs a week then realized i needed to focus on school

15-25

full time - 50-60 hours/week

average 20

Yes, hours per week varied per semester. It was in the range of 20-30 hours per week.

22 hours

No

N/A

Yes, UIC Campus Housing Peer Mentor/ Senior Peer Mentor 15-20 hours

no

yes, not part-time but full time, often 50hrs a week while commuting. School was the hardest thing I have ever done.

(8.) On average, how much many hours did you spend on each ChE course outside of class?

about 2-4 hours

At least 20

lots! 40-50

35 hours per week

5-10

40 hours a week.

Probably 40 hours a week.

About 10/week

A lot! Especially in the last 2 years.

25 hours

20

~40

8 hours a week

about 6 hours per week per course

how would i know? In all honesty this question is odd id say 5 hours per week maybe

10-15/weeeek

Around 7-8 hours for every class a week

7/week

20-30

around 10 hours

On average, I spent around 10-15 hours a week on each course outside of class.No

10 hours

15hours/week per course

probably 2-5 hours per week, depending on which class.

6-7

4/day

10 hours a day

a lot

3-12 hours per class depending on the difficulty.

15 hours

5hrs a week

everyday i would spend 3 hours at the minimum a day.

30

about 2-3 hours a day

few hours daily

Heavily depends on course, at least 2 hours for every credit of the course (5 credit course = 10 hours outside of class)

On average, each course involved about 5-10 hours per week

0-15 hours

35

5-6 hours a week

3-4 hr/day

20-30

12

5 hours/week

5 per week

About 20-25 hours per week was typical

3-7 hours a class per week

5 hours

N/A

15+ atleast

3 hours a day

about 90 hours if I make the average over all ChE courses.

(9.) Did you participate in any summer research/internship program at UIC or at other schools? Did you complete an internship at a company? What company? Was this a useful experience? What specifically did you learn that was most useful?

No, I am trying to find an intern before graduate.

Research from 2014 to graduation. Internship at Rema Foods Inc. The internship gave me an insight on the possibilities a chemical engineer would have in the food industry and helped me make my decision for my future. The research gave me the opportunity to think in a different way and learn by doing.

research. with sharma. no internships

I worked as an intern in SiNode Systems company where I learned how to make silicon-graphene electrode materials for lithium-ion batteries.

I had a summer internship at WEC Energy group. It was not very useful for me because my mentor did not know how to properly use an intern. It would have been helpful in understanding how to work in an office setting, but I already had experience with that.

I completed an internship at WEC Energy, it was for the natural gas division and I learned a lot. I learned how the job world functions.

I did research at UIC through GPIIP. It was very useful because I got to experience working with others and learning new procedures in a lab.

Yes I interned at ITW Security & Brand Identity as well as superior graphite. It was a very useful experience. I was able to apply much of what I learned. It also helped me really focus in on what matters vs what is just academic jargon.

No

I have only completed research at UIC under faculty at the CHEME department. This summer i hope to find another internship.

research at uic

Yes, I participated in research for 2 semesters and had an internship at Aluminum Anodizing Corporation. I would consider both useful. Research was useful because it gave me more of an insight on Graduate School and I learned about my research topic which I was already interested in. My internship was useful because I got to see what Engineer in a quality department does day to day.

no

I completed an internship at the Stepan Company over the summer of 2016. It got extended to a co-op that I am continuing right now as well and will continue it until I graduate in May.

yes; yes: Center for Neighborhood Technology, Gelita USA Chicago. Very useful. be willing to help, punctual and tuned in.

Yes/Yes/Terracon/Yes/Professional Etiquette

No

I researched with Professor Sharma during the summer of 2016. The most useful thing I learned was how to identify and solve unexpected problems independently.

no I just continued my part time job

none

No

Yes. FESP at UIC, Sustainability internship at UIC, IT-Business Analyst Intern at Mars, Inc. and Intern for AthenaVisual, Inc.

N/A

I participated in summer internships at Integrys Energy Group and Cooper Standard. I completed a summer research experience for undergraduates at the NSF Engineering Research Center for Biorenewable Chemicals. The experience at Cooper Standard was the most useful since I was able to work in a manufacturing environment and the experience helped me get a job.

Yes, an internship at Peoples Gas. It was very useful in seeing what the actual practice of an engineer is. I learned a lot about the natural gas industry and processing.

Yes; No; N/A

no

Honor's College research experience, at the bioengineering department, Applied Soft matter Summer Research Opportunity with Dr. Sharma. Very useful experiences. I plan to continue to graduate school so it gave me a good research background.

I did not

No

None

Summer research with Dr. Berry's lab. No internships. They need to be pushed more to the students.

No not yet

Research for Dr. Kral

Girls Who Code internship - taught young girls programming, great experience, learned more about coding & career development

No I did not.

Yes. Summer research with Dr. Akpa and internship with Gelita. Yes, very helpful. I learned how an industrial plant works and how I can apply my engineering skills.

yes, SunCoke Energy. It was great, I got to apply all of my classes to the real world

I interned at DS Smith Rapak. The experience was invaluable. Working in a professional environment, managing my own time as well as other occasionally, and ways to present data in a meaningful way

Yes No N/A Yes What the research environment is like, and the skills required to "survive" in that environment.

no

yes sustainability and research assistant

no because i have a full time job as a chemist

research, really enjoyed it

Summer research at Millikin University. Internship at Evapco Midwest. It was a very useful experience, as it showed me what the job dynamic is typically like.

Not yet still applying

Yes, with UIC. It was very useful and allowed me to experience some problems engineers have to deal with regarding equipment, research, and safety

N/A

Yes, ASSURE Program at UIC within Dr Vivek Sharma's ODES Lab Yes, Underwriters Laboratories, yes, politics, lean six sigma, intersectionality of engineering fields

summer research at UIC

I did not, I worked full time and felt some misplaced loyalty.

(10.) Have you found a job? If so, in which company and industry? Please indicate your job title or specialization.

No

No.

no

not yet

WEC Energy - Gas Engineer.

Yes. I will be working at FM Global as a Field Engineer.

Not yet.

Not yet

Not yet, looking for internships since i will not be graduating till next December.

no

No

no

no.

no i am currently searching.

No

No

I am going to be working for UOP in the field of gas processing and hydrogen.

no

none

No

Not yet

Not yet!

Not yet no.

I will accept the manufacturing engineering professional development program position at Northrop Grumman.

Yes, Peoples Gas: Associate Engineer in Engineering Design

No

no

NA

No

No

None

Not yet.

No

no

No, have not looked

No

No.

yes, Consultant Engineer

Not yet

No

not yet

no

workiung on it

not yet

none

Not yet.

No

No

N/A

No

no

I have not so far. I only started searching a week ago.

(11.) Are you planning to attend a graduate or a professional school? If so please list the name of the school and the specific program (M.S./Ph.D/M.D.) that you will enroll in, or those programs that you have looked into.

Not at this moment

Potentially. I will be applying to UIC for a PhD or DePaul University for MBA.

considering it

No

No, if I do attend grad school it would be for business.

No

NO

No

No

yes, possibly a phd but not sure where yet

No

maybe U of I (Ph.D)

no.

yes. I don't know yet.

No

Not currently

N/A

no

no

No

Yes, MBA

Not immediately no but it is a possibility.

No

Not right now, but yes is the future. M.S but not sure what school

No

no

I am still in the process of applying. Selected schools are: CCNY, Columbia, university of Maryland College park, Tufts university, UCLA, and USC

No

No

No graduate school

Not unless I work for a company that will help me get my MBA or MPM

No

have not decided. Thinking of graduate school

Ph.D programs at Columbia, NYU, UC Berkeley, Northwestern, UMinnesota

No

Not decided yet.

n/a

I plan on doing so after working for a few years. I am currently looking at the Energy Engineering Masters Program at UIC.

Yes UIC (M.S.) University Austin (Ph.D) Berkeley (M.S.) MSU (M.S.) University of Minnesota (M.S.)

not at the moment

get a MBA but that in like 5 years

possibly energy engineering at uic

I would like to obtain an MPM in the future.

still deciding

I would like to attend graduate school. My first choice is Purdue for a M.S. in pharmaceutical engineering.

N/A

Not right now, but in the future maybe

N/A

I would love to, however not immediately Berkley Logic and The Methodology of Science Chemical Engineering at CalTech

not planned for the upcoming year

I would like to one day but in the meantime I am in too much debt.

(12.) Do you have a strategy in mind for continued professional development and/or continue learning? For example, do you intend to take the FE/PE exam at some point in your career? If so, when?

If the company that I join in the future required then I will take the PE exam

I hope to eventually branch out to the managerial side of engineering companies. I will do so by taking classes independently and potentially going for an MBA.

no

I'm planning to take the FE exam after I get my degree.

I will be taking the FE next year. I think it should be something that is discussed about during junior year because the types of materials we talk about junior year are the questions that are on the FE test.

Not sure if I will. If I do, I want to take it not too long after graduation.

Plan on taking the FE in the spring. Also plan on getting Six Sigma Green belt in the spring.

I intend to find a job and start working

I plan on pursuing higher education but not sure exactly when.

not sure

I intend on taking the exam within the next year

FE, MBA, Masters in Energy and PE

FE yes and probably PE... FE maybe next year.

Mehh, Ill probably take the FE right after graduation.

I plan to take the PE exam in 6 or 7 years into my carrer

I would like to take the FE exam at some point very soon after I graduate, and at some point I will take advantage of UOP's sponsorship of continued learning and graduate degrees.

not yet

Take the FE and PE soon as i can.

Although I am not attending graduate school next year, I do intend to eventually get a Masters degree.

I do plan on taking the FE, probably after a year or two of working

Yes, within a year or two of graduating.

I intend to return to school to obtain an MBA or a masters in Engineering Management. I intend to take the FE exam in the near future, and the PE exam sometime in the future.

I am taking the FE and PE exams as soon as possible.

No

no

I plan of taking the FE immediately after graduation.

I intend to take the FE before graduating and the PE when I meet the requirements.

FE exam next semester

Hopefully i can get a job and eventually return to school sometime in the future.

FE should be encouraged prepared for in undergraduate program. Maybe part of 499. Definitely going take the PE in the future.

NO

I do plan on taking exams. Have not decided when

FE right after graduation maybe, otherwise have not heavily considered

I would like to eventually go back to get my MBA.

Intend to take the FE Exam.

n/a

I plan on fleshing out my CS abilities with my ChE knowledge. I will be most likely take the FE at some point, but I have not decided on when.

Continue membership of AIChE and eventually I do want to take the FE/PE, ideally with a year or two of graduation.

no

no

FE in the spring

I would like to take the FE exam sooner rather than later.

yes after first couple years

I would like to take the PE exam when I am officially done with my schooling.

I plan on taking the PE exam after I have a career that I am satisfied and am well established in.

I will like to take the FE exam right after graduation.

N/A

Yes yes, hopefully upon employment in industry, post graduation I would like to pursue education in some fashion throughout the rest of my life. Varying contexts/lenses enrich one's learning and I hope to pursue learning within a corporation, graduate school, organization, etc. Becoming a professor is something I would like to do as well. Teachers have continuously inspired me and have helped me sculpt myself into the person I am today. I hope to do the same for others.

I plan to go to graduate school within a few years

I absolutely want to continue developing professionally. I could do this by networking, attending conferences, and furthering my education. I would like to take the PE exam but that could not happen until 2021

Valid receipt codes for ChE exit:

12723
14798
18153
18488
18613
19107
19413
19658
22992
23166
24334
24857
27833
29080
30739

33858
33887
35352
38595
39516
39949
40174
40190
40214
41601
41942
42012
42749
43246
44508
44991
45126
47744
47753
51082
51178
52225
54771
55959
57516
59164
60245
61352
64486
68864
72787
73563
73598
74648
75225
75632
79906
81494
83403
84257
85579